

Battery Sizing for an Off-Grid Solar + Wind System: What We Found (V2)

The question

If you're building a solar + wind power system with no backup generator and no grid connection, how big a battery do you need so the system reliably meets demand? We compared two ways to answer that:

1. **The physics-based method:** simulate the system hour-by-hour for a full year, for every possible battery size, and pick the smallest one that keeps the lights on 99% of the time.
2. **The AI method:** train a neural network on thousands of these physics-based answers, then ask it to predict the right battery size directly for a new system, without re-running the year-long simulation.

This is a from-scratch redo ("V2") of an earlier version of this project, rebuilt to match a specific course's exact engineering formulas, focused tightly on this one comparison (no other AI models muddying the picture), and switched to studying a fully **off-grid** system rather than one with a diesel generator backing it up.

Why off-grid, and what almost went wrong

Studying "off-grid" means reliability is a hard requirement: if the battery can't cover a bad week, the system fails, full stop — there's no generator to fall back on. That's the harder, more useful test of whether the AI has really learned the right lesson.

But there's a trap: if you don't also make sure most of the systems you're studying have *enough* solar and wind capacity relative to their demand, almost none of them will be able to meet that reliability requirement no matter how big a battery you install — a battery only smooths out short gaps, it can't manufacture energy that was never generated in the first place. We checked this before committing to the full study: an initial test showed only 39 out of 100 sample systems could actually meet the reliability target off-grid. We fixed the mix of systems being studied (shrinking the typical electricity demand relative to the solar/wind capacity already being tested) and re-checked twice more before running the full study — landing at 88% and 90% of sample systems being solvable, comfortably enough to learn from, while still keeping a genuine ~10% minority of unsolvable cases so the results stay honest rather than artificially perfect.

The full-scale study (5,000 example systems) came in at **90.5% solvable**, confirming the fix worked.

What we found

The AI is very accurate. Across ten separate tests (each time retraining on a different random slice of the data), the AI's battery-size predictions matched the physics-based answer to within about 163 kWh on average, out of typical battery sizes in the thousands of kWh — a 99.4% statistical fit. This wasn't a lucky one-off: repeating the test ten times with different data splits gave nearly identical results every time.

More training data helps — up to a point. Accuracy improved sharply as we gave the AI more example systems to learn from, but that improvement essentially stopped after about 2,000 examples. Giving it the full available ~3,200 barely helped at all. If we wanted a meaningfully better model in the future, more of the same kind of data isn't the answer — better information about each system would be.

The AI learned the right lessons, not lucky guesses. We tested which pieces of information the AI relies on most by scrambling each one and seeing how much its accuracy dropped. The top factors were exactly what an engineer would expect to matter for an off-grid system: how much the solar and wind output swings from month to month, how bad the single worst month is, and how strict the reliability requirement is. That's reassuring — it means the AI is reasoning about the right things.

But trusting the AI blindly still carries real risk. We took the AI's predicted battery sizes, rounded them up to the nearest installable battery unit (never down), and actually re-simulated each system for a full year to see if it really held up. **94% did.** The other 6% fell short of the reliability target even after rounding up — meaning if you installed exactly what the AI suggested without double-checking, about 1 in 17 systems would come up short during a bad stretch of weather. This is the main practical takeaway: **the AI is a fast, very good first estimate, but a physics-based double-check before actually building the system is still the responsible final step**, not an optional extra.

Battery price matters, but doesn't change the recommended size.

Testing battery prices from 220 to 400 EUR per kWh, the total system cost per unit of electricity delivered ranged from about 0.36 to 0.46 EUR/kWh — battery cost alone explains a third to nearly half of total system cost. Notably, changing the assumed battery price does *not* change which battery size gets recommended off-grid — that decision is driven entirely by the reliability requirement, not by cost, since there's no cheaper "good enough" option when reliability is a hard requirement.

The AI really is much faster — but how much faster depends on how you use it. Running the full physics-based simulation takes about a third of a second per system. If you ask the AI to evaluate many systems at once (the realistic case for exploring lots of design options quickly), it's about **2,500 times faster** per system. If you ask it to evaluate just one system on its own, it's still faster — about **6.5 times** — but the speed advantage shrinks a lot, because most of that single request's time is just the overhead of asking the question at all, not the actual computation. Either way, none of this includes the time it took to train the AI in the first place (a few minutes, done once) — the speed payoff only shows up once you're evaluating many new systems using that already-trained model.

Bottom line

For quickly narrowing down battery size across many candidate off-grid system designs, the AI model is fast and reliably close to the physics-based answer. For finalizing an actual system that will be built, the physics-based check is still necessary — about 1 in 17 of the AI's rounded-up recommendations would not have met the reliability target without it. Used together — AI for fast exploration, physics-based simulation for final confirmation — this gives the best of both: speed during design, and confidence before construction.

A follow-on question: what if we add a backup generator?

Everything above describes a fully off-grid system — no backup generator, so the battery has to be big enough to survive every bad week on its own. That's why the solar and wind panels had to be built so much larger than the actual demand: about 60% of all the power they generated over the year was simply thrown away (curtailed), because there was nowhere to put it once the battery was full.

As a follow-on check, we turned the diesel backup generator on and increased the electricity demand so it's better matched to what the solar and wind can produce. With a generator to fall back on, the system doesn't need a battery big enough to survive on its own — it just needs a battery big enough to be the *cheapest* combination of battery and generator fuel.

The results: with a 24-battery-unit system (6,000 kWh), solar and wind cover **74%** of the electricity used over the year, the generator covers the other 26%, and far less power goes to waste — down from 60% curtailed to 18%. We also checked, battery size by battery size, where the sweet spot is: **adding battery is worth it up to about 74% renewable power.** Below that point, every bit of battery you add saves more in generator fuel than it costs. Beyond that point, it flips — the last bit of battery capacity mostly sits idle waiting for rare, extreme weeks, and costs more than the fuel it saves. Pushing renewable coverage from 74% up toward 80% this way would cost

about 30% more per unit of electricity than stopping at the sweet spot, for only a small gain in renewable share.

This is a separate, smaller check — it did not retrain the AI model or regenerate the full 5,000-system dataset described above, both of which remain valid results of the off-grid study. It answers a different, narrower question: for a hybrid system with a generator as backup, how much battery is actually worth paying for.